

Example 1

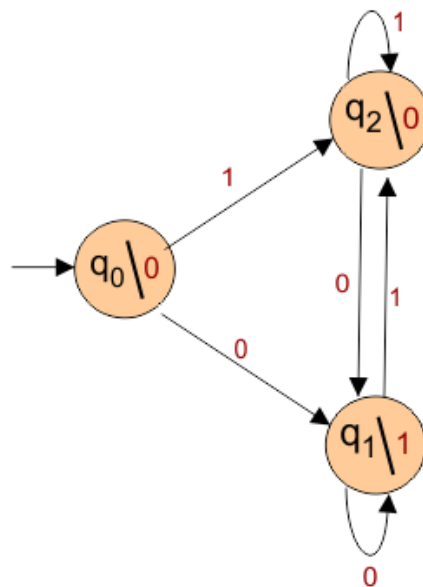
Design a Moore machine to generate 1's complement of a given binary number.

Solution

According to first complement, if the given input is "1" then the output will be "0" and if the input is "0" then the output will be "1".

For First complement there should be at least three states.

- One state is start state.
 - The 2nd state is to accept "1's" as input and produces output as "0".
 - The 3rd state will accept "0's" as input and producing output as "1".
- Hence the final Moore machine of given example 1 will be,



For instance, take one binary number 111001 then

Input		1	1	1	0	0	1
State	q0	q1	q1	q1	q2	q0	q1
Output	a	0	0	0	1	1	0

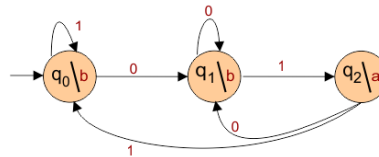
Thus we get 0000110 as 1's complement of 111001, we can neglect the initial 0 and the output which we get is 000110 which is 1's complement of 111001.

Example 02

Construct a Automata Moore Machine that prints “a” whenever the sequence “01” is encountered in any input binary string.

Solution

Automata Moore Machine for given example is given below



Design a Moore machine for a binary input sequence such that if it has a substring 101, the machine outputs A, if the input has substring 110, it outputs B otherwise it outputs C.

Solution

For designing such a machine, we will check two conditions, and those are 101 and 110. If we get 101, the output will be A, and if we recognize 110, the output will be B. For other strings, the output will be C.

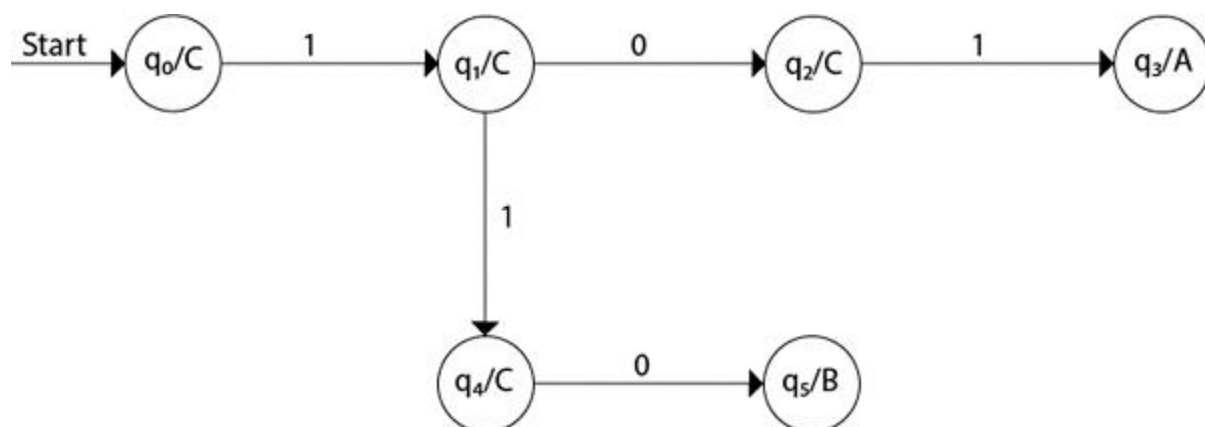
Moore machine has 6 tuples

$(Q, q_0, \Sigma, O, \delta, \lambda)$

Where,

- Q : Finite set of states
- q_0 : Initial state of machine
- Σ : Finite set of input symbols
- O : Output alphabet
- δ : Transition function where $Q \times \Sigma \rightarrow Q$
- λ : Output function where $Q \rightarrow O$

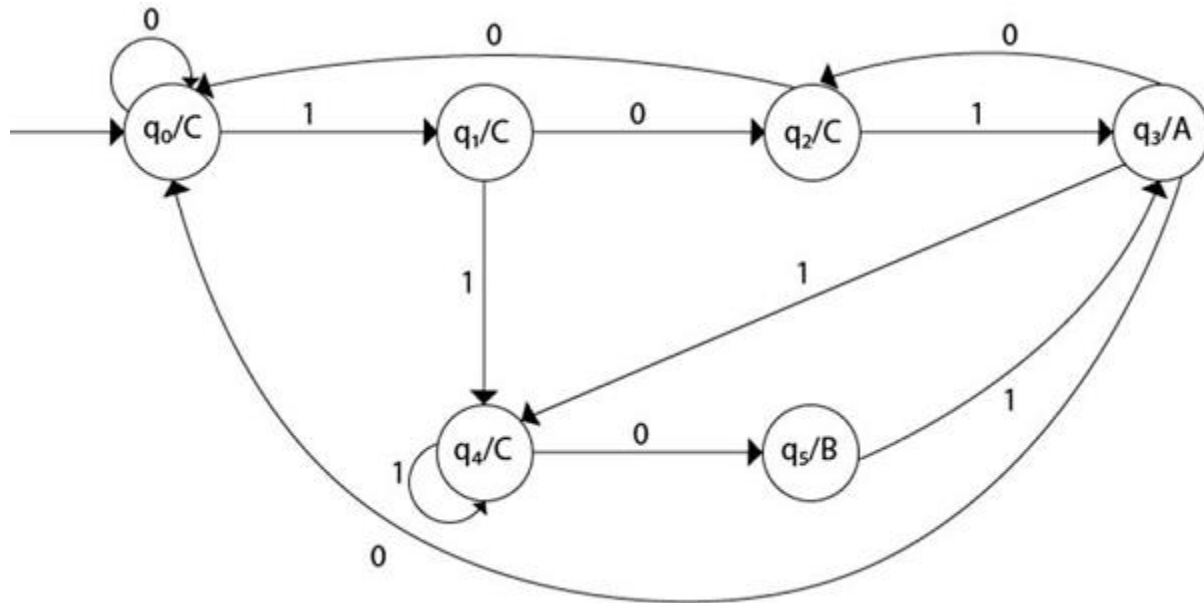
The partial diagram will be as follows –



Example 1

Now, we will insert the possibilities of 0's and 1's for each state.

Thus, the Moore machine becomes as follows –



Explanation

- **Step 1** – q_0 is the start state on input '0' goes to q_0 and on '1' goes to q_1 generating output C.
- **Step 2** – q_1 on '0' goes to q_2 and on '1' goes to q_4 generating output C.
- **Step 3** – q_2 on '0' goes to q_0 and on '1' goes to q_3 generating output C.
- **Step 4** – q_3 on input '0' goes to q_2 and to q_0 and on '1' goes to q_4 generating output A.
- **Step 5** – q_4 on input '0' goes to q_5 and on '1' goes to q_4 generating output C.
- **Step 6** – q_5 on input 1 goes to q_3 generating an output B.

Example 2

Construct a Moore machine On input string bababbb, the output is 01100100.

Input alphabet – $\Sigma = \{a, b\}$

Output alphabet – $\Gamma = \{0, 1\}$

States – q_0, q_1, q_2, q_3

State Transition table

The transition table is as follows –

Current State	a	b	Output
q0	q3	q2	0
q1	q1	q0	0
q2	q2	q3	1
q3	q0	q1	0

State Transition diagram

The transition diagram is as follows –

